"How many times have you had an intuition that could not be rapidly verified?"

This seminar is about that gap.

From Research Question(s) to Article ToC

AI as a tool for scientific thinking: a personal workflow

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Why Using AI in Research Question Formulation?

"There is no best AI —
only the one most aligned
with your problem."

Core idea

AI does **not replace** the researcher.

It works in a **coupled architecture**:
human (and machine) in the loop together.

Structural analogy

The real strength of AI lies in
hierarchical representations
— not in "creativity".

Pattern abstraction.
Structural similarity detection.
Cross-domain transfer.

The Human-in-the-Loop Protocol

*Not damage control — **preventive design**.*

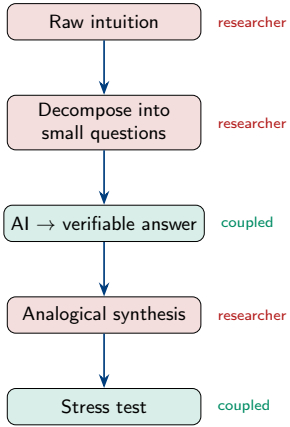


Figure: The Human In The Loop (HITL) protocol

Two Modes of Error Detection

Mode 1 — Technical (Raman code)

Small code chunks, one at a time.

Error signal: NaN values, constant correlations across spectra, anomalous plots.

Visible to anyone who looks at the output.

Error is localised because the question is localised.

Mode 2 — Conceptual (semantic invariant)

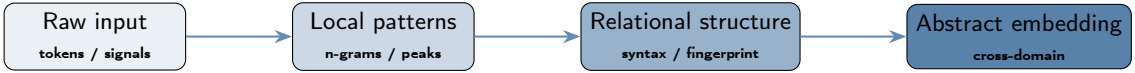
First version used the **universal set** of objects instead of the category closure under the functor.

Not wrong — but **too strong** for the intended claim.

Only visible to someone who knows the domain.

	Technical	**Conceptual**
Detected by	anomalous output	semantic misalignment
Visible to	anyone	domain expert only
Correction	technical	intentional

From Tokens to Structured Representations



Core idea

The AI works at the **abstract embedding** level —
where structural similarity across domains becomes visible.

This is why the same model can help a physicist,
a biologist, and a computational linguist.

Essential structure(s)

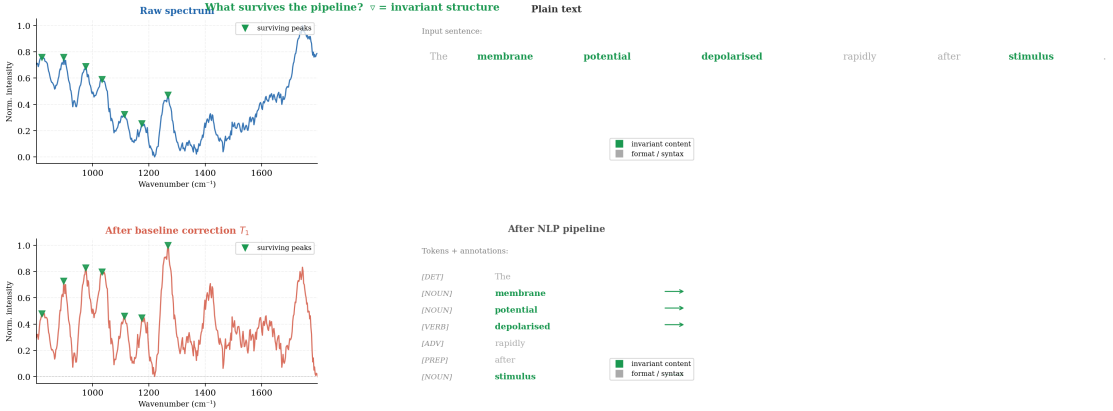
*What is the **essential structure***
that survives a series of transformations?

Core idea

This analogy formalises the concept of **invariant**
as the essential structure preserved across a pipeline
$T_1 \circ T_2 \circ \cdots \circ T_n$ —
independently of the domain.

Spectroscopy · NLP pipelines · Category theory

Analogy 1 — What Survives the processing?



Analogy 1 — The First, Rough Research Question

"Given a processing pipeline

$$T_1 \circ T_2 \circ \cdots \circ T_n,$$

*which **essential structure** of the raw input survives?*
*And how do we **measure** survival?"*

World 1: Spectroscopy

Raw spectrum $\xrightarrow{T_1,\ldots,T_n}$ processed

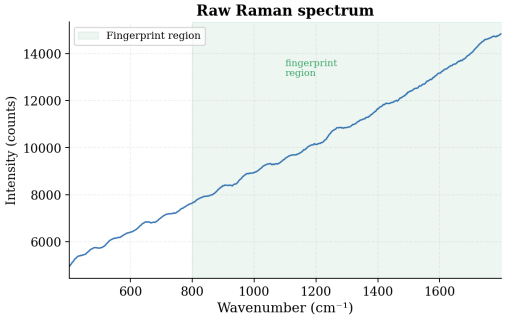
Essential structure = **chemical fingerprint** survives denoising, baseline, normalisation

World 2: NLP

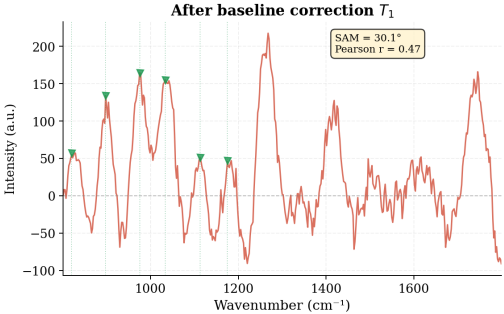
Raw text $\xrightarrow{f_1,\ldots,f_n}$ enriched

Essential structure = **semantic content** survives format conversion, annotation, parsing

Analogy 1 — The Question Emerges from the Data



Raw spectrum



After baseline correction T_1

Core idea

"After all these transformations — denoising, baseline removal, normalisation — ***what is still there?*** *And how do we know it survived?"*

Analogy 1 — First Dialogue with the AI

Prompt A — Generic question

"I am processing Raman spectra through a pipeline of transformations (denoising, baseline correction, normalisation). Can you suggest metrics to measure whether the chemical invariants of the raw spectrum are preserved after each step?"

AI response — selected excerpt

*"Several metrics can quantify spectral invariance across transformations. Global measures include **Spectral Angle Mapper (SAM)** and **Pearson correlation** on the fingerprint region. For peak-level analysis, consider **peak position shift**, **prominence ratio**, and **area ratio** between raw and processed spectra. For domain-specific validation, a **band hit-rate** against known reference peaks would directly measure chemical fingerprint preservation."*

Analogy 1 — Refining: Small Questions, Verifiable Answers

Prompt B — Specific

"For each known diagnostic band, I want to check: (1) is the peak present after processing? (2) has its position shifted? (3) has its prominence changed? Weight the result by band importance (reference / core / desired / secondary)."

Prompt C — Verification

"Run this on spectrum X18, raw vs baseline-corrected. Show me the band-by-band result."

Output — immediately verifiable:

Band	Present	Shift
975 cm^{-1}	✓	$< 1\,\mathrm{cm}^{-1}$
1033 cm^{-1}	✓	$< 1\,\mathrm{cm}^{-1}$
1268 cm^{-1}	✓	$< 1\,\mathrm{cm}^{-1}$
1418 cm^{-1}	✓	$< 1\,\mathrm{cm}^{-1}$
1745 cm^{-1}	✓	$< 1\,\mathrm{cm}^{-1}$
13 / 15 bands survive T_1		

Core idea

Each answer is **immediately checkable** on the plot — no blind trust.
This is the Human-in-the-Loop protocol.

Analogy 1 — The Same Pattern in NLP

Prompt A — NLP side

"I have a pipeline of NLP tools applied to raw text (tokenisation, NER, format conversion). Can you suggest how to verify that the semantic content of the original text is preserved at each step? Think in terms of measurable, verifiable checks."

AI response — selected excerpt

*"To verify semantic preservation, use external lexical resources as reference anchors: **WordNet** synsets to check that word senses are stable across transformations; **ontologies** (e.g. UMLS, DBpedia) to verify that named entities retain their semantic class; **WSD tools** to detect sense shifts introduced by the pipeline. A step is suspect if the sense distribution of key terms changes significantly."*

Analogy 1 — What Survives the Conversion?

Raw text — XML

```
<sentence>
The membrane potential
depolarised after stimulus.
<ref doi="10.1093/brain/
awz001"/>
</sentence>
```

$\xrightarrow{F}$

format converter

Processed text — JSON

```
{
"text": "The membrane
potential depolarised
after stimulus.",
"entities": [{"type":"PROC",
"value":"depolarised"}]
}
// doi reference: lost
```

Element	Survives?	Check
membrane (WordNet)	✓	synset stable
potential (WSD)	✓	sense: physics
depolarised (UMLS)	✓	concept preserved
stimulus (WordNet)	✓	synset stable
doi reference	×	**lost**

Critical note

F preserves **semantic content** but **forgets** para-textual structure (external references, metadata).

Is this loss acceptable?
It depends on what I must preserve.

Analogy 1 — The Formal Mapping

Structural analogy

Spectroscopy $\longleftrightarrow$ NLP pipelines

Common mechanism: the **forgetful functor**

Spectroscopy		**NLP / Category Theory**
Raw spectrum	$\leftrightarrow$	Document $D \in \mathrm{Ob}(\mathcal{C}_1)$
Processing step T_i	$\leftrightarrow$	NLP tool / morphism $t : D \to D'$
Pipeline $T_1 \circ \cdots \circ T_n$	$\leftrightarrow$	Composed morphisms in $\mathcal{C}$
Chemical fingerprint	$\leftrightarrow$	Semantic content $\mathcal{S}$
Invariant functor I	$\leftrightarrow$	$I_i : \mathcal{C}_i \to \mathcal{S}$
Admissible T_i	$\leftrightarrow$	$I(\mathrm{output}) = I(\mathrm{input})$

*An admissible processing step is one that **preserves the invariant by construction** — not just for tested cases.*

Analogy 1 — The Dialogue Converges

Prompt — Cross-domain

"In Raman spectroscopy, the chemical fingerprint is the essential structure that survives a pipeline $T_1 \circ \cdots \circ T_n$. In NLP, semantic content survives format conversion. In both cases something is **forgotten** and something is **preserved**. Can you formalise this so the definition holds in both domains simultaneously?"

AI response — excerpt

*"Both cases instantiate the same structure: a **forgetful functor** $F : \mathcal{C}_1 \to \mathcal{C}_2$ that discards representation-specific information while preserving essential structure via an **invariant functor** $I : \mathcal{C} \to \mathcal{S}$.*

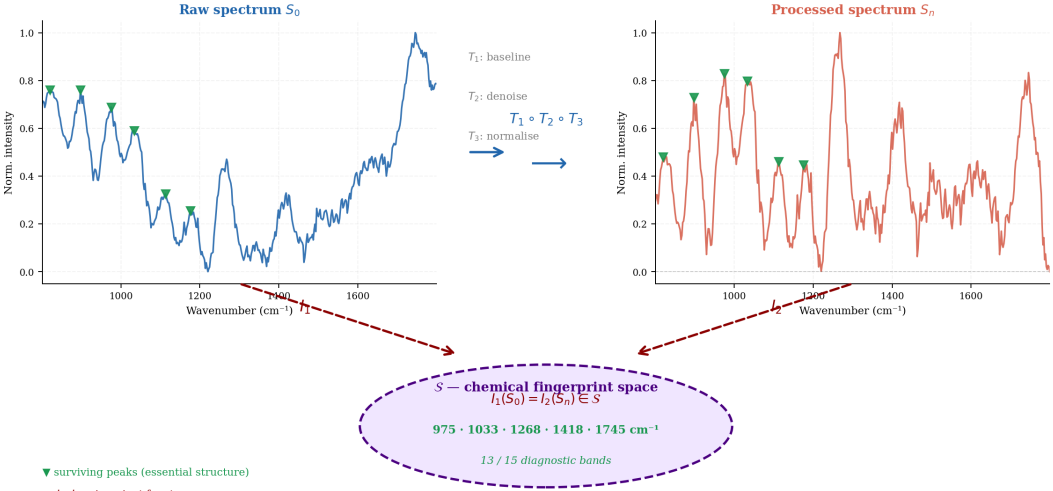
*A transformation is **admissible** iff $I(\text{output}) = I(\text{input})$. What is forgotten is not lost by accident — it is what I is **defined not to see**."*

Core idea

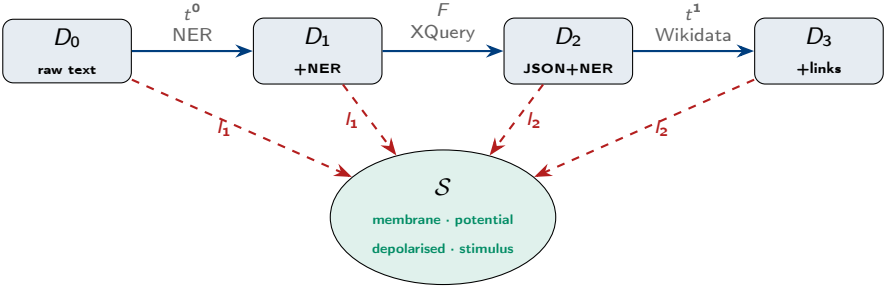
The AI names what we were already doing:

Spectroscopy:	F forgets baseline & noise	I preserves peaks
NLP:	F forgets format & metadata	I preserves semantics

Analogy 1 — The Spectroscopy Pipeline



Analogy 1 — The Same Structure in NLP



Core idea

$\mathcal{S}$ contains what **all representations share**:
the essential structure — independent of format.

`doi:10.1093/brain/awz001` is **not** in $\mathcal{S}$ — F forgets it. *By design.*

Analogy 1 — The Common Theoretical Foundation

$$\begin{tikzcd}
\mathcal{C}_1 \arrow[rr, "F"] \arrow[dr, "I_1"'] & & \mathcal{C}_2 \arrow[dl, "I_2"] \\
& \mathcal{S} &
\end{tikzcd}$$

$I_1 = I_2 \circ F$

$\mathcal{C}_1$	raw spectra / raw text
$\mathcal{C}_2$	processed spectra / JSON
F	pipeline / converter
$\mathcal{S}$	fingerprint / semantics (*essential structure*)

Why a functor, not a function?

A **function** maps objects:
$f : A \to B$

A **functor** maps objects **and transformations**:
$F : \mathcal{C}_1 \to \mathcal{C}_2$

— coherently: if $t \circ t'$ in $\mathcal{C}_1$
then $F(t) \circ F(t') = F(t \circ t')$

"A function tells you where you arrive.
A functor tells you the path is the same."

Analogy 1 — Where Does It Break?

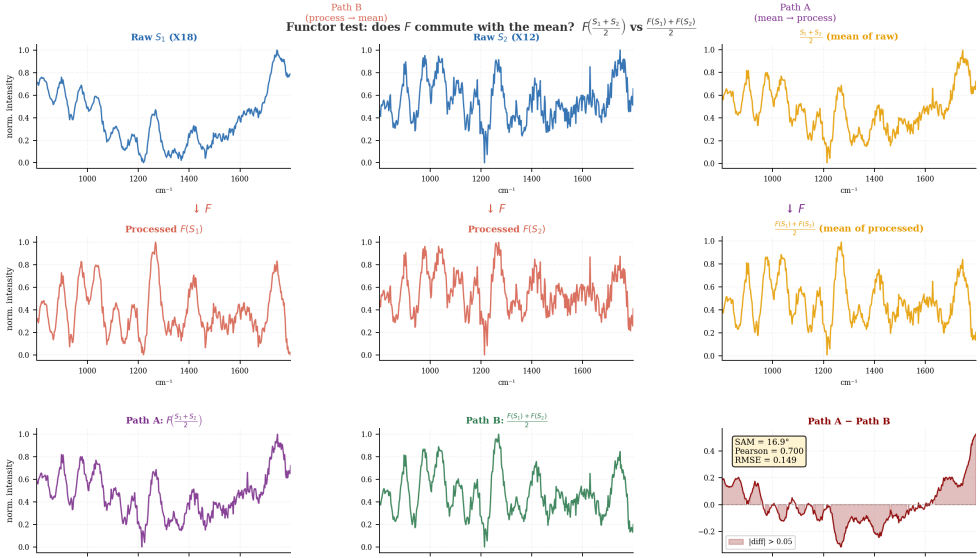
Prompt C — Stress test

"Where does this analogy break down? Under what conditions do the chemical fingerprint and lexical semantics behave fundamentally differently with respect to the invariant concept? What are the limits of the formalisation?"

Critical note

- In spectroscopy, I is **measurable** with explicit metrics (e.g. spectral similarity, peak preservation)
  — in NLP, I is often **only implicitly defined**
- Pipeline transformations in spectroscopy are **physically constrained**
  — NLP tools have much wider degrees of freedom
- **Metrics**: which ones reliably detect invariant loss?
  (HolmesCapScore, SAM — work in progress)

Analogy 1 — The Functor Test



Analogy 1 — An Open Invitation

Core idea

The condition $I_1 = I_2 \circ F$ is **mathematically falsifiable**.

An admissible step is correct **by construction** — not just for tested cases.

Where does this structure appear?

Raman spectroscopy pipelines
NLP / document processing
Image processing pipelines
Genomic sequencing workflows
Neural signal processing
Your domain?

Open questions:

- Can admissibility be verified automatically?
- Does the forgetful spectrum extend to other signal domains?

HolmesCapScore

A combined top-down / bottom-up metric: counts **how many** diagnostic peaks survive, weighted by importance tier.

"Three clues make a proof"
— Agatha Christie
(not Sherlock Holmes — verified with AI)

Analogy 1 — The Forgetful Functor: Scaling Difficulty

	Raman	NLP	Genomics	Neural signals
$\mathcal{S}$ fixed?	✓ physics	∼ task-dependent	∼ task-dependent	× non-stationary
Admissibility	✓ HolmesCapScore	∼ WordNet / WSD	∼ GATK practices	× ICA unstable
Forgetting controlled?	✓ physically constrained	∼ by design (doi example)	∼ partially constrained	× often implicit
Functor test	✓ SAM = 16.9°	? open	? open	? open

Critical note

The structure is there in all four domains.

The difficulty scales with how well $\mathcal{S}$ can be defined.

Raman: physics constrains $\mathcal{S}$ — and we can **test** it.
NLP: $\mathcal{S}$ is definable — forgetting is **by design** (doi example).
Genomics / Neural: $\mathcal{S}$ is task-dependent or non-stationary.

This difficulty is not a flaw — it is a domain-specific open question.

References: Analogy 1

- Ferrari, A. C. **and** D. M. Basko (2007). "Raman spectroscopy as a versatile tool for studying the properties of graphene". **in** *Nature Nanotechnology*: 2, **pages** 235–253. DOI: 10.1038/nnano.2007.308.
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- Mikolov, Tomas, Wen-tau Yih **and** Geoffrey Zweig (2013). "Linguistic Regularities in Continuous Space Word Representations". **in** *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*: **pages** 746–751.
- Spivak, David I. (2014). *Category Theory for the Sciences*. MIT Press. DOI: 10.7551/mitpress/9612.001.0001.

From Question to TOC — Analogy 1

Initial intuition

1. These two pipelines look similar
2. Something survives the transformations
3. Can we measure what is preserved?

$\Rightarrow$

TOC after AI dialogue

1. Research question: what survives $T_1 \circ \cdots \circ T_n$?
2. Formal mapping: forgetful functor F, invariant I, admissibility $I_1 = I_2 \circ F$
3. Operationalisation: HolmesCapScore and SAM as functor tests
4. Stress test: $\mathcal{S}$ definability scales across domains
5. Open question: when does F commute with averaging?

Core idea

The AI named what we were already doing:
"What is forgotten is not lost by accident — it is what we is ***defined not to see****."*

Questions?

Analogy 2 — Generalised Lotka-Volterra & Sense Shifting

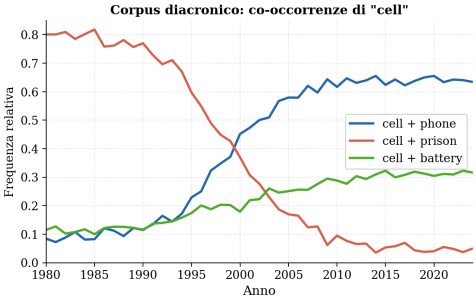
*Can the **dynamics of competing species** describe the **evolution of word meanings**?*

Core idea

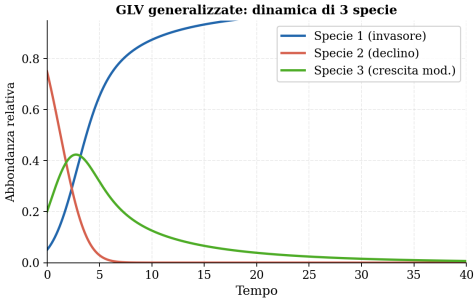
This analogy formalises **semantic shift** as a selective dynamic among competing word senses — using the same equations that describe species competition in ecology.

Ecology · Computational linguistics · Evolutionary game theory

Analogy 2 — When Ecology and Linguistics Look Like the Same Thing



Diachronic corpus: co-occurrences of *"cell"*
Simulated data illustrating a real pattern (sigmoid model + Gaussian noise)



Generalised Lotka-Volterra: 3 competing species

Core idea

"Two plots. Two fields. One question."

Analogy 2 — The First, Rough Research Question

"If two plots from distant worlds
look this similar,

is there a common theoretical substrate?"

World 1: Ecology

Species competing for resources
→ dynamics of relative abundances

World 2: Linguistics

Word senses competing for usage
→ dynamics of diachronic co-occurrences

This is the raw version of the research question.
*The pattern is there, the AI **named** it.*

Analogy 2 — Dialogue with the AI (1/3)

Prompt A — The starting point

"A colleague presented diachronic co-occurrence data for a target word — counting collocates across decades and using KLD to measure semantic shift. Can you explain the rationale and show a minimal example?"

AI response

"KLD measures how much the collocate distribution of a word has shifted between two periods. If p is the 1990s distribution and q the 2020s distribution over collocates $w_1, \ldots, w_n$:

$$D_{KL}(q \| p) = \sum_i q_i \log \frac{q_i}{p_i}$$

A large value signals a semantic shift; near zero signals stability. The asymmetry is intentional: it measures how surprising is q from the perspective of p."

Analogy 2 — What KLD Tells Us (and What It Does Not)

AI response — on the word *cell*

	p_{90s}	q_{20s}	$\log \frac{q}{p}$
phone	0.077	0.640	+2.12
prison	0.769	0.040	−2.96
battery	0.154	0.320	+0.73

$$D_{KL}(q\|p) = \sum_i q_i \log \frac{q_i}{p_i}$$

$$= \underbrace{0.640 \cdot 2.12}_{\textit{phone}} + \underbrace{0.040 \cdot (-2.96)}_{\textit{prison}} + \underbrace{0.320 \cdot 0.73}_{\textit{battery}} = 1.47 \text{ nats}$$

$$D_{JS}(p,q) = \tfrac{1}{2} D_{KL}(p\|M) + \tfrac{1}{2} D_{KL}(q\|M) = 0.34 \text{ nats}, \quad M = \tfrac{p+q}{2}$$

$$\frac{D_{JS}}{\log 2} = 49\% \text{ of maximum possible shift}$$

Core idea

What this answers:
The meaning of *cell* has shifted substantially — nearly half the maximum possible divergence.

Critical note

What this cannot answer:
Which sense won? Which went extinct? At what rate? Through what process?

KLD is a **distance between snapshots**. The plot suggested a **trajectory**.

Analogy 2 — Dialogue with the AI (2/3)

Prompt B — Pushing further

"Interesting. Are there other strategies for measuring semantic shift? What about embedding-based approaches?"

AI response

Method	Type	Limitation
KLD / JSD	Statistical	Measures *how much*, not *how*
Aligned word2vec	Embedding	Needs careful alignment
BERT-based	Contextual	Computationally expensive
Graph analysis	Structural	Requires graph modeling

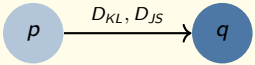
Critical note

All methods measure a **distance between states**.
None describes the **process** visible in the plot.

Analogy 2 — Static Measures vs Dynamic Models

Static approach

Two snapshots: p (1990s) and q (2020s).



Answers: *how far apart are the two states?*

Lost: the path, the rate, the interactions.

Dynamic approach

A trajectory: $p(t)$ evolving continuously.

Answers: *how does the system evolve?*

Gained: rates, interactions, attractors.

A **dynamical system** is a rule that assigns to every state $p(t)$ a velocity $\dot{p}(t)$:

$$\dot{p}_i = F_i(p_1, \ldots, p_n)$$

The plot suggested that semantic shift has a velocity.

Analogy 2 — A Special Case: Competitive Dynamics

Generic dynamical system

$$\dot{p}_i = F_i(p_1, \ldots, p_n)$$

Each p_i evolves according to some rule F_i — possibly independent.

No constraint on interactions.

Competitive dynamics

$$\dot{p}_i = p_i \cdot g_i(p_1, \ldots, p_n)$$

Growth rate g_i depends on **all other competitors**.

The system is coupled: no sense evolves in isolation.

Why competition?

Word senses share a finite resource: **collected corpora**.

If *phone* grows, it must do so at the expense of *prison* or *battery*.

This is not an assumption — it follows from $\sum_i p_i = 1$.

Ecology → Linguistics

Species compete for **resources**.
Word senses compete for **usage**.

Same constraint: $\sum_i p_i = 1$.
Same structure: coupled ODE.

Analogy 2 — Dialogue with the AI (3/3)

Prompt C — The creative leap

"What if semantic shift were modelled as a replicator competition? Each word sense competes for usage, with fitness estimated from observed frequency shifts."

AI response

"This is a well-posed reformulation. In replicator dynamics, each sense i has fitness f_i and frequency $p_i(t)$. The replicator equation reads:

$$\dot{p}_i = p_i\left(f_i - \bar{f}\right), \quad \bar{f} = \sum_j p_j f_j$$

*For $n > 2$ competing senses, the natural generalisation is the **Generalised Lotka-Volterra** system — which also encodes pairwise interactions between senses, not just selection against the mean fitness."*

Core idea

The researcher brings the idea. The AI names the framework: **GLV**.

Analogy 2 — The Formal Mapping

Structural analogy

Ecology $\longleftrightarrow$ Computational Linguistics

GLV (ecology)		**Diachronic corpus**
Species i	$\leftrightarrow$	Collocate / word sense
Relative abundance x_i	$\leftrightarrow$	Relative frequency of use
Fitness f_i	$\leftrightarrow$	Log growth rate
Mean fitness $\bar{f}$	$\leftrightarrow$	Average semantic drift
Extinction	$\leftrightarrow$	Obsolescence of a sense
Invasion	$\leftrightarrow$	Emergence of a new meaning

Analogy 2 — From GLV to Replicator: Formal Relation

Generalised Lotka-Volterra:

$$\dot{x}_i = x_i \left(r_i + \sum_{j=1}^{n} a_{ij} x_j \right) \qquad i = 1, \ldots, n$$

Replicator equation (special case restricted to the n-dimensional simplex $\sum_i x_i = 1$):

$$\dot{x}_i = x_i \left(f_i(\mathrm{x}) - \bar{f}(\mathrm{x}) \right), \qquad \bar{f}(\mathrm{x}) = \sum_{j=1}^{n} x_j f_j(\mathrm{x})$$

Core idea

We are not using something *similar* to GLV — we are using a **formally exact special case**. The simplex constraint $\sum_i x_i = 1$ models a **closed vocabulary**: total usage is conserved.

Special case $n = 2$: reduces to **logistic growth** $\dot{x} = x(1-x)(f_1 - f_2)$

Analogy 2 — From GLV to Replicator — Sketched Derivation

Start from GLV (absolute abundances x_i):

$$\dot{x}_i = x_i \left(r_i + \sum_j a_{ij}\, x_j \right) = x_i\, f_i(\mathrm{x})$$

Define total mass and relative frequencies $S(t) = \sum_k x_k, \qquad \nu_i = \dfrac{x_i}{S} \quad \Rightarrow \quad \sum_i \nu_i = 1$

Differentiate $\dot{\nu}_i = \dfrac{\dot{x}_i S - x_i \dot{S}}{S^2}$. Since $\dot{S} = \sum_k \dot{x}_k = \sum_k x_k f_k = S\bar{f}, \qquad \bar{f}(\mathrm{x}) = \sum_k \nu_k\, f_k(\mathrm{x}),$

we obtain $\dot{\nu}_i = \dfrac{x_i}{S}\big(f_i(\mathrm{x}) - \bar{f}(\mathrm{x})\big) = \nu_i\,\big(f_i(\mathrm{x}) - \bar{f}(\mathrm{x})\big).$

Core idea

The replicator equation is GLV written in **relative coordinates**. The subtraction of $\bar{f}$ arises from the dynamics of total mass $S(t)$. In the semantic model: $\nu_i(t) \to x_i(t)$ once restricted to the simplex.

Analogy 2 — Notation at a Glance

Symbol	Meaning	Origin
$x_i(t)$	relative frequency of sense i at time t	replicator model (simplex)
$f_i(\mathrm{x})$	fitness of sense i — state-dependent	replicator / GLV
$\bar{f}(\mathrm{x}) = \sum_j x_j f_j$	mean fitness of the system	replicator / GLV
p_i	observed frequency of sense i — 1990s snapshot	corpus data
q_i	observed frequency of sense i — 2020s snapshot	corpus data
$\hat{f}_i = \log \frac{q_i}{p_i}$	zeroth-order scalar proxy for $f_i(\mathrm{x})$ ($T = 1$: one unit = observation period)	estimated from data
$\bar{f} = \sum_i p_i \hat{f}_i$	mean fitness — weighted by 1990s snapshot	estimated from data

Critical note

$x_i(t)$ and p_i, q_i are **not the same object**: $x_i(t)$ is the dynamical variable; p_i, q_i are static corpus snapshots used to initialise the model.

Analogy 2 — Estimating Fitness from Corpus Data

Model assumption: each sense i grows exponentially:

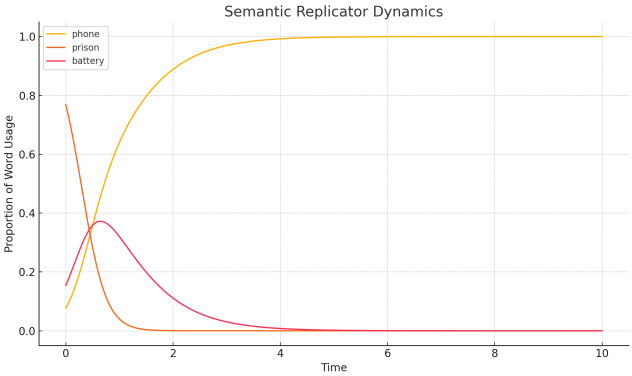
$$x_i(t) = x_i(0)\, e^{\hat{f}_i\, t} \quad \Longrightarrow \quad \hat{f}_i = \log \frac{q_i}{p_i}$$

Time is measured in units of the observation period (one unit = 1990s → 2020s), so $T = 1$ by convention.

$\hat{f}_i$ is a **zeroth-order scalar proxy** for the state-dependent fitness $f_i(\mathrm{x}(t))$ of the replicator equation: it assumes constant selection pressure and $a_{ij} = 0$.

Collocate	p (1990s)	q (2020s)	$\hat{f}_i = \log(q/p)$	$\hat{f}_i - \bar{f}$
phone	0.077	0.640	$+2.12$	$+4.12$
prison	0.769	0.040	-2.96	-0.96
battery	0.154	0.320	$+0.73$	$+2.73$
$\bar{f} = \sum_i p_i \hat{f}_i$			-2.00	0.00

Analogy 2 — Semantic Replicator Dynamics: *cell*



Replicator ODE integration Real corpus data show the same qualitative trend of simulated data

Starting from the 1990s distribution, the model integrates forward in time.

Result:

- *phone* dominates ($f \approx +4.12$)
- *prison* collapses ($f \approx -0.96$)
- *battery* grows moderately before collapsing ($f \approx +2.73$)

Semantic shift of *cell*:
prison $\rightarrow$ **technology**

Analogy 2 — Where Does It Break?

Prompt C — Stress test

"Which properties of GLV do not transfer to the linguistic corpus? GLV assumes continuous, differentiable populations — a corpus is discrete and sparse. Does this break the analogy or make it more interesting?"

Critical note

- Corpus is **discrete and sparse**: fitness is *estimated*, not directly observed — assumes two snapshots suffice
- The **interaction matrix** a_{ij} between collocates is rarely identifiable empirically — assumes $a_{ij} = 0$ in our model
- The linguistic simplex is **not closed**: new collocates emerge (unpredicted invasions) — assumes fixed vocabulary
- Fitness is **non-stationary**: it changes with historical context — assumes $\hat{f}_i$ constant over the period

References: Analogy 2

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From Question to TOC — Analogy 2

Initial intuition

1. These two plots look similar
2. Maybe ecology describes language?
3. Show the analogy, compute something

$\Rightarrow$

TOC after AI dialogue

1. Visual analogy $\rightarrow$ formal question
2. GLV $\rightarrow$ replicator: exact derivation
3. Data mapping + fitness estimation
4. Stress test: where the analogy breaks
5. Open problem: estimating a_{ij}

Core idea

The AI did not generate ideas.
It forced the logical structure to become explicit — transforming a visual intuition into a **falsifiable argument**.

Questions?

Analogy 3 — A Question from Far Away

"Can we interpret the minimisation of a loss function as a geodesic on a (Riemannian) manifold?"

This analogy is different

No visual starting point.

The intuition comes from a **conceptual question**,
not from two similar plots.

The bridge

Mechanics $\longleftrightarrow$ **Learning**

Least action principle
$\longleftrightarrow$
Loss minimisation

Analogy 3 — The Loss: What We Are Minimising

Regression — MSE:

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \|y_i - \hat{y}_i(\theta)\|^2$$

Classification — Cross-Entropy:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{n} y_i \log p_\theta(y_i|x_i)$$

General — Negative Log-Likelihood:

$$\mathcal{L}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} \log p_\theta(y_i|x_i)$$

Core idea

All three share the same structure:

$$\mathcal{L}(\theta) = \mathbb{E}_{\text{data}}\left[\ell(y,\, p_\theta(\,\cdot\,|x))\right]$$

The model output $\hat{y} = f_\theta(x)$ defines a **conditional distribution**:

$$p_\theta(y|x) \propto \exp\left(-\frac{\|y - f_\theta(x)\|^2}{2\sigma^2}\right)$$

MSE = NLL under Gaussian noise.
Cross-entropy = NLL under categorical model.
Same object — different assumptions.

Analogy 3 — The Gradient Depends on the Metric

1. Standard gradient descent (Euclidean):

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t)$$

Assumes all directions in parameter space are equivalent.

The parameter space has a natural metric — the **Fisher information matrix**:

$$g_{ij}(\theta) = \mathbb{E}[\partial_i \log p_\theta \cdot \partial_j \log p_\theta]$$

Measures how distinguishable two nearby distributions p_θ and $p_{\theta+d\theta}$ are.

2. Natural gradient (Amari):

$$\theta_{t+1} = \theta_t - \eta\, g^{-1}(\theta_t) \nabla_\theta \mathcal{L}(\theta_t)$$

Steepest descent in the intrinsic geometry of $\mathcal{M}$.

Core idea

The same loss $\mathcal{L}$ — two different geometries.
Standard GD treats $\mathcal{M}$ as flat. Natural gradient respects its curvature.

What does it mean to move on a curved parameter space? — next slide.

Analogy 3 — The Training Trajectory as a Curve on $\mathcal{M}$

The natural gradient defines a **vector field** on $\mathcal{M}$:

$$\dot{\theta}^i(t) = -g^{ij}(\theta)\,\partial_j \mathcal{L}(\theta) \qquad \Leftrightarrow \qquad \dot{\theta}(t) = -\operatorname{grad}_g \mathcal{L}(\theta)$$

What is grad_g?

In Euclidean space, $\nabla\mathcal{L}$ points in the direction of steepest ascent — all directions treated equally.

On $\mathcal{M}$, $\operatorname{grad}_g \mathcal{L}$ is the same concept but **weighted by the local geometry**: directions where the distribution changes rapidly (high Fisher curvature) are penalised.

The training trajectory $\theta(t)$ is the integral curve of this vector field — the unique curve that, at every point, follows the direction prescribed by $\operatorname{grad}_g \mathcal{L}$.

Core idea

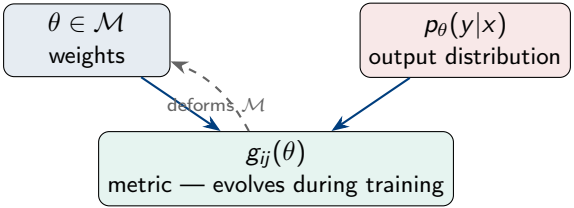
Formally: given an initial condition $\theta(0) = \theta_0$, the trajectory solves: $\dot{\theta}(t) = -\operatorname{grad}_g \mathcal{L}(\theta(t)), \qquad \theta(0) = \theta_0$ This is a **Cauchy problem on $\mathcal{M}$** — a differential equation with initial condition, whose solution exists locally and is unique (under standard regularity conditions).

Analogy 3 — The Manifold $\mathcal{M}$

Core idea

The parameter space is **not flat**.

The weights θ live on a Riemannian manifold $\mathcal{M}$ whose metric $g_{ij}(\theta)$ depends on both the weights **and** the output distributions $p_\theta(y|x)$.



The data shapes p_θ, which determines g_{ij}, which deforms $\mathcal{M}$, which changes how θ moves.
A coupled system.

Analogy 3 — The Strategy

We want to interpret training as geometry.
Here is the path.

1. Define geodesic on $\mathcal{M}$ — *Beyond Fisher metric*

↓

2. Define two distinct curves on M — *Integral vs. Geodesic*

↓

3. Curvature of $\mathcal{M}$ = Curvature of Loss — *Connect the Hessian to the metric tensor and introduce the Ricci tensor*

↓

4. The Ricci tensor and the curvature — *Ricci and curvature*

↓

5. The conjecture: The metric g_{ij} evolves: conjecture $\partial_\tau g_{ij} = -2R_{ij}$ — *the open conjecture*

Figure: Our strategy

Analogy 3 — The Geodesic: Shortest Path on $\mathcal{M}$

The shortest path between two points on $\mathcal{M}$ minimises the arc length:

$$\delta \int_a^b \sqrt{g_{ij}(\theta)\,\dot{\theta}^i \dot{\theta}^j}\; dt = 0$$

Applying the Euler-Lagrange equations gives the **geodesic equation**:

$$\ddot{\theta}^k + \Gamma^k_{ij}(\theta)\,\dot{\theta}^i \dot{\theta}^j = 0$$

What are the Christoffel symbols $\Gamma^k_{ij}(\theta)$?

$$\Gamma^k_{ij}(\theta) = \tfrac{1}{2}\, g^{kl}(\theta)\,(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

They measure **how the metric changes** from point to point on $\mathcal{M}$.

$\Gamma^k_{ij}(\theta) = 0$ everywhere $\Leftrightarrow$ $\mathcal{M}$ is flat $\Leftrightarrow$ standard GD is exact.
$\Gamma^k_{ij}(\theta) \neq 0$ $\Leftrightarrow$ the geometry deflects the trajectory — standard GD makes an error.

Note: $g_{ij}(\theta)$ and $\Gamma^k_{ij}(\theta)$ both depend on θ — they change at every point along the training trajectory. The object $p_\theta(y|x)$ is what generates this dependence: it shapes g_{ij}, which shapes Γ^k_{ij}, which shapes the geodesic.

Analogy 3 — Two Curves on $\mathcal{M}$

Training trajectory

Forced curve — driven by $\mathcal{L}$:

$$\dot{\theta}^i = -g^{ij}(\theta)\,\partial_j \mathcal{L}(\theta)$$

External force: the loss gradient.

Standard GD: sets $g^{ij} = \delta^{ij}$ (ignores curvature). Natural GD: uses $g^{ij}(\theta)$ (first-order correction only).

Geodesic

Free curve — geometry only:

$$\ddot{\theta}^k + \Gamma^k_{ij}(\theta)\,\dot{\theta}^i\dot{\theta}^j = 0$$

No external force. The curve follows the curvature of $\mathcal{M}$ — like a free particle on a curved space (second-order equation, richer than gradient flow).

Core idea

These are two different objects on $\mathcal{M}$. The training trajectory is **pushed** by $\mathcal{L}$. The geodesic is **free** — shaped only by $g_{ij}(\theta)$.

The central question of Analogy 3: does there exist a metric $g^*_{ij}(\theta)$ such that the two curves coincide?

If yes: the loss is not a force — it is the geometry itself.

Analogy 3 — Curvature of $\mathcal{M}$ = Curvature of the Loss

The key identity (Fisher, under model distribution):

$$g_{ij}(\theta) \;=\; \mathbb{E}[\partial_i \log p_\theta \cdot \partial_j \log p_\theta] \;\approx\; \partial_i \partial_j \mathcal{L}(\theta) \;=\; H_{ij}(\theta)$$

Both $g_{ij}(\theta)$ and $H_{ij}(\theta)$ **depend on** θ — they change at every point of the training trajectory.

Curvature of $\mathcal{M}$: the Ricci tensor

$$R_{ij}(\theta) = \partial_k \Gamma^k_{ij} - \partial_j \Gamma^k_{ik} + \Gamma^k_{kl}\Gamma^l_{ij} - \Gamma^k_{jl}\Gamma^l_{ik}$$

$R_{ij}(\theta)$ depends on θ through $g_{ij}(\theta)$ and its derivatives.

Intuition: R_{ij} measures how fast g_{ij} changes — i.e. how fast the Hessian of the loss changes.

Core idea

High $R_{ij}(\theta)$ = rapidly varying H_{ij} = sharp, ill-conditioned region of the loss landscape.

Low $R_{ij}(\theta)$ = slowly varying H_{ij} = flat, well-conditioned region.

The geometry of $\mathcal{M}$ is the geometry of the loss.

Critical note

$g_{ij} = H_{ij}$ exactly only for NLL under the model distribution. For finite datasets: approximation. Both objects depend on θ — they are not constants.

Analogy 3 — Ricci Flow: Selective Curvature Redistribution

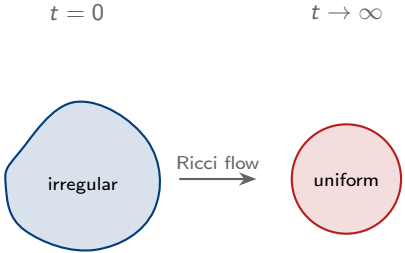
$$\frac{\partial g_{ij}(\theta)}{\partial t} = -2\, R_{ij}(\theta)$$

The flow acts **selectively** on $\mathcal{M}$:

- $R_{ij}(\theta) > 0$: region **contracts**
  sharp loss — ill-conditioned parameters
- $R_{ij}(\theta) < 0$: region **expands**
  flat loss — well-conditioned parameters
- $R_{ij}(\theta) = 0$: **unchanged**

Both $g_{ij}(\theta)$ and $R_{ij}(\theta)$ depend on θ — the flow deforms the manifold point by point, driven by the local loss landscape.

Used by Perelman (2003) to prove the Poincaré conjecture: any simply-connected 3-manifold flows to a sphere.



Core idea

Convergence = uniform curvature
= uniform Hessian of the loss
= well-conditioned optimisation
everywhere on $\mathcal{M}$.

Analogy 3 — The Conjecture

Conjecture

There exists a reparametrisation $\tau(\theta(t))$ of training time such that the Fisher metric evolves as a Ricci flow **along the training trajectory**:

$$\frac{\partial g_{ij}(\theta)}{\partial \tau} = -2\, R_{ij}(\theta)$$

Why this is not the standard Ricci flow:

- Standard Ricci flow (Hamilton 1982): the metric evolves **autonomously** in the space of metrics — no trajectory, no data.
- Here: the metric evolves **along** $\theta(t)$, driven by the data through $p_\theta(y|x)$. A weaker statement — but precise and falsifiable.

Core idea

If true: training does not just minimise $\mathcal{L}$ — it **uniformises the curvature** of the loss landscape, redistributing conditioning along the trajectory.

Convergence = the training trajectory has exhausted the curvature of $\mathcal{M}$. No numerical evidence yet — this is a conjecture.

Analogy 3 — The Open Question

Does there exist a metric $g^*_{ij}(\theta)$ such that the geodesic on $\mathcal{M}$ coincides with the loss-minimising path?

If yes

The loss $\mathcal{L}(\theta)$ is not an external force.

It is **encoded in the geometry** of $\mathcal{M}$ through $g^*_{ij}(\theta)$.

Training is a **free geodesic** — no force, only curvature.

Analogy with GR: gravity is not a force, it is spacetime curvature.
Here: the loss is not a force, it is parameter space curvature.

Critical note

What needs to be proved:

- Existence of $g^*_{ij}(\theta)$
- Regularity conditions on $\mathcal{M}$
- Connection between g^*_{ij} and the Ricci flow conjecture

Analogy 3 — What the Dialogue Produced

This question emerged from the AI dialogue.
It was not there before the session.

Core idea

The AI did not answer the question.

It helped **formulate it precisely** —
turning a vague intuition about geodesics and loss
into a falsifiable mathematical statement.

Potential single-author paper.

This is the human-in-the-loop protocol at its limit:
not verification of answers —
*but **generation of questions**.*

References: Analogy 3

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From Question to TOC — Analogy 3

Initial intuition

1. Loss minimisation looks like a physical trajectory
2. Maybe gradient descent is geometry?
3. Explore the connection informally

$\Rightarrow$

TOC after AI dialogue

1. Geodesic equation vs. standard gradient descent
2. Conjecture: training as Ricci flow on $\mathcal{M}$
3. Open question: does the loss-minimising geodesic exist?

Core idea

The AI dialogue turned a metaphor into a **open conjecture** with a precise mathematical statement — potentially a single-author paper.

From Research Question(s) to Article TOC

AI as a tool for scientific thinking: a personal workflow

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Repository	`https://github.com/riccardodg/FromRQ2ToC`
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Thanks

Questions?

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From Research Question(s) to Article TOC

AI as a tool for scientific thinking: a personal workflow

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Analogy 3 Bonus Toy Model — A Minimal Network

Values chosen deliberately: class 1 dominant, class 3 almost absent.

Architecture:

- 2 inputs $x = (x_1, x_2)$, 3 classes, 1 layer
- 9 parameters: $W \in \mathbb{R}^{3\times 2}$, $b \in \mathbb{R}^3$

Pre-activations:

$$z_k = \sum_j w_{kj} x_j + b_k$$

Softmax: $p_k = \dfrac{e^{z_k}}{\sum_m e^{z_m}}$

Our values: $x = (1, 2), \quad y_{\text{true}} = (1, 0, 0)$

$$W = \begin{pmatrix} 1.0 & 0.5 \\ -0.5 & 1.0 \\ 0.2 & -0.8 \end{pmatrix}, \quad b = \begin{pmatrix} 0.1 \\ -0.2 \\ 0.3 \end{pmatrix}$$

Result:

$$z = (2.10,\ 1.30,\ -1.10)$$

$$p = (0.671,\ 0.301,\ 0.027)$$

Critical note

$p_1 = 0.671$ — class 1 **dominates;** $p_3 = 0.027$ — class 3 **almost absent.**

This asymmetry is the point — the geometry will reflect it.

Analogy 3 Bonus Toy Model — Energy, Loss, Gradient

Loss used: cross-entropy (NLL)

Neuron energies $E_k = -\log p_k$:

$$E_1 = -\log(0.671) = 0.399$$
$$E_2 = -\log(0.301) = 1.199$$
$$E_3 = -\log(0.027) = 3.599$$

High energy = low probability = surprise

Loss — true class $y_{\text{true}} = 1$, so we use E_1:
(only the true class contributes to NLL)

$$\mathcal{L} = -\log p_1 = E_1 = 0.399$$

Gradient — explicit calculation:

$$\frac{\partial \mathcal{L}}{\partial w_{kj}} = \underbrace{(p_k - y_k)}_{\delta_k} \cdot x_j$$

$$\delta = p - y = \begin{pmatrix} 0.671 \\ 0.301 \\ 0.027 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -0.329 \\ +0.301 \\ +0.027 \end{pmatrix}$$

param	δ_k	x_j	grad $\delta_k x_j$
w_{11}	-0.329	1	-0.329
w_{12}	-0.329	2	-0.658
w_{21}	$+0.301$	1	$+0.301$
w_{22}	$+0.301$	2	$+0.603$
w_{31}	$+0.027$	1	$+0.027$
w_{32}	$+0.027$	2	$+0.055$

Analogy 3 Bonus Toy Model — Fisher Metric & Natural Gradient

The geometry knows which directions are underrepresented.

Fisher diagonal $g_{w_{kj},w_{kj}} = p_k(1-p_k)\,x_j^2$:

param	g_{ii}	param	g_{ii}	param	g_{ii}
w_{11}	0.221	w_{21}	0.211	w_{31}	0.027
w_{12}	0.883	w_{22}	0.843	w_{32}	0.106
b_1	0.221	b_2	0.211	b_3	0.027

Std vs Natural gradient $\tilde{\nabla} = g^{-1}\nabla\mathcal{L}$:

param	std	natural	ratio
w_{11}	-0.329	-0.166	$0.50\times$
w_{12}	-0.658	-0.331	$0.50\times$
w_{21}	$+0.301$	$+0.152$	$0.50\times$
w_{22}	$+0.603$	$+0.304$	$0.50\times$
w_{31}	$+0.027$	$+0.083$	$3.03\times$
w_{32}	$+0.055$	$+0.166$	$3.03\times$

Core idea

$p_3 = 0.027$

$\downarrow$

$g_{w_{31}} = p_3(1-p_3) \approx p_3$

$g_{w_{31}} = 0.027$

$\downarrow$

g^{-1} large

$\downarrow$

natural gradient **amplifies** $(3\times)$

low prob = low Fisher = large correction

When $p_k \ll 1$, $g_{w_{kj}} \approx p_k\,x_j^2$

The metric encodes the class distribution.

Analogy 3 Bonus Toy Model — Gradient Derivation

Dense slide — shown for completeness.

Starting from:

$$\mathcal{L}(\theta) = -\sum_k y_k \log p_k, \qquad p_k = \frac{e^{z_k}}{\sum_m e^{z_m}}, \qquad z_k = \sum_j w_{kj} x_j + b_k$$

Step 1 — derivative of softmax:

$$\frac{\partial p_k}{\partial z_l} = p_k(\delta_{kl} - p_l) \qquad \begin{cases} k = l: & p_k(1 - p_k) \\ k \neq l: & -p_k p_l \end{cases}$$

Step 2 — chain rule on $\mathcal{L}$:

$$\frac{\partial \mathcal{L}}{\partial z_l} = -\sum_k y_k \frac{\partial \log p_k}{\partial z_l} = -\sum_k \frac{y_k}{p_k} \cdot p_k(\delta_{kl} - p_l) = -\sum_k y_k(\delta_{kl} - p_l) = -y_l + p_l \underbrace{\sum_k y_k}_{=1} = p_l - y_l = \delta_l$$

Step 3 — chain rule on w_{lj}: $\frac{\partial \mathcal{L}}{\partial w_{lj}} = \frac{\partial \mathcal{L}}{\partial z_l} \cdot \frac{\partial z_l}{\partial w_{lj}} = \delta_l \cdot x_j$ ■

Bonus — Derivation: $E(\theta)$ and the Three Losses

1. NLL — exact by definition:

$$E_i = -\log p_\theta(y_i|x_i) \quad \Rightarrow \quad E(\theta) = \sum_i E_i = -\sum_i \log p_\theta(y_i|x_i) = n\,\mathcal{L}_{\text{NLL}}(\theta)$$

2. Cross-entropy — exact (special case of NLL):

$$p_\theta(y|x) = \prod_k p_k^{y_k} \quad \Rightarrow \quad -\log p_\theta = -\sum_k y_k \log p_k = \mathcal{L}_{\text{CE}}$$

$$\Rightarrow \quad E(\theta) = n\,\mathcal{L}_{\text{CE}}(\theta) \qquad \text{exact}$$

3. MSE — under Gaussian assumption:

$$p_\theta(y|x) = \mathcal{N}(f_\theta(x),\, \sigma^2 I) \quad \Rightarrow \quad -\log p_\theta(y|x) = \frac{\|y - f_\theta(x)\|^2}{2\sigma^2} + \underbrace{\frac{d}{2}\log(2\pi\sigma^2)}_{\text{const w.r.t. } \theta}$$

$$\Rightarrow \quad E_i = \frac{\|y_i - \hat{y}_i\|^2}{2\sigma^2} + C \quad \Rightarrow \quad E(\theta) = \frac{n}{2\sigma^2}\,\mathcal{L}_{\text{MSE}}(\theta) + nC$$

exact up to scaling and additive constant

The additive constant nC does not affect the minimum of $\mathcal{L}$ nor the geometry of $\mathcal{M}$.

Bonus — A Gas of Neurons

What kind of "gas" are the neurons?

Gas model	Neuron model	Regime
Ideal gas	Independent neurons (no correlations)	High temperature (random init)
Mean field / Ising	Each neuron feels average of others	Trained network (Hopfield limit)
Beyond Ising (3+ body)	Multi-body interactions (non-symmetric)	Attention, deep networks

Core idea

Training = gas evolves from **disordered** (high T, random) to **structured** (low T, correlated).
The loss is the **thermodynamic potential** being minimised.
Training is a phase transition.

Analogy 3 — Energy of a Single Neuron

What is the "energy" of a neuron?

A neuron carries **information** about its input. The natural measure is:

$$E_i = -\log p_i \qquad \text{(surprisal)}$$

p_i = probability that neuron i produces its current output given the input ($p_i = e^{z_i} / \sum_k e^{z_k}$, *softmax*)

Taylor expansion around p_i^0 (operating point = current p_i before perturbation δp_i):

$$E_i \approx \underbrace{-\log p_i^0}_{\text{0th}} + \underbrace{\frac{\delta p_i}{p_i^0}}_{\text{1st}} + \underbrace{\frac{(\delta p_i)^2}{2(p_i^0)^2}}_{\text{2nd}} + \mathcal{O}(3)$$

0th order: $-\log p_i^0$

Baseline energy. Constant w.r.t. perturbation.
Summed over neurons → current **NLL**.

1st order: $\delta p_i / p_i^0$

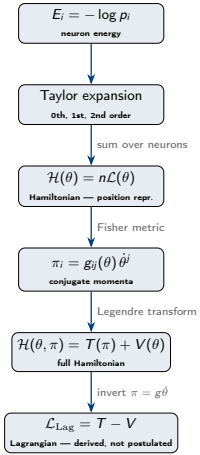
Linear response to perturbation.
Summed over neurons → **loss gradient**.

2nd order: $(\delta p_i)^2 / 2(p_i^0)^2$

Quadratic curvature term.
Summed over neurons → **Fisher metric** g_{ij}.

Analogy 3 — From Neuron Energy to Lagrangian Mechanics

Everything emerges from a single object: $E_i = -\log p_i$



Analogy 3 — Step 2: The Network Hamiltonian

Sum over all neurons (N total):

$$\mathcal{H}(\theta) = \sum_{i=1}^{N} E_i(\theta) = -\sum_{i=1}^{N} \log p_i(\theta) = n\,\mathcal{L}(\theta)$$

Core idea

The loss $\mathcal{L}(\theta)$ **is** the Hamiltonian of the network — in position representation.

Minimising $\mathcal{L}$ = finding the **ground state** of a statistical mechanical system whose energy is information surprise.

From Taylor — three objects

0th order	$\to \mathcal{L}(\theta)$ = energy
1st order	$\to \nabla\mathcal{L}$ = force
2nd order	$\to g_{ij}(\theta)$ = inertia

Critical note

$\mathcal{H}(\theta)$ depends only on θ — no momenta yet. This is **not** the full Hamiltonian. The momenta come from the metric.

Analogy 3 — Step 3: Conjugate Momenta from the Fisher Metric

In classical mechanics, the conjugate momentum to a coordinate q^i is $p_i = \partial \mathcal{L}_{\mathrm{Lag}} / \partial \dot{q}^i$.

On $\mathcal{M}$ with metric $g_{ij}(\theta)$, the natural kinetic term is $T = \frac{1}{2} g_{ij} \dot{\theta}^i \dot{\theta}^j$, which gives:

$$\pi_i = \frac{\partial T}{\partial \dot{\theta}^i} = g_{ij}(\theta)\, \dot{\theta}^j$$

Physical meaning of π_i

π_i is the **momentum of the parameter** θ^i weighted by the local geometry of $\mathcal{M}$.

In flat space ($g_{ij} = \delta_{ij}$): $\pi_i = \dot{\theta}^i$ — ordinary velocity.
On curved $\mathcal{M}$: directions of high Fisher curvature carry **more momentum** for the same velocity — like moving through a denser medium.

Core idea

The Fisher matrix $g_{ij}(\theta)$ is the **inertia tensor** of the parameter space.
It determines how much "resistance" each direction of $\mathcal{M}$ opposes to changes in θ — exactly what the natural gradient corrects for.

Analogy 3 — Step 4: The Full Hamiltonian via Legendre Transform

Inverting $\pi_i = g_{ij}\dot{\theta}^j$ gives $\dot{\theta}^i = g^{ij}\pi_j$. The **Legendre transform** of T is:

$$\mathcal{H}(\theta,\pi) = \pi_i\,\dot{\theta}^i - T = g^{ij}\pi_i\pi_j - \tfrac{1}{2}g^{ij}\pi_i\pi_j = \underbrace{\tfrac{1}{2}g^{ij}(\theta)\,\pi_i\pi_j}_{T(\theta,\pi)} + \underbrace{n\,\mathcal{L}(\theta)}_{V(\theta)}$$

Critical note

This resolves the T-V tension.

In $(\theta,\dot{\theta})$ space: $T = \frac{1}{2}g_{ij}\dot{\theta}^i\dot{\theta}^j$ depends on $g_{ij}(\theta)$ — so T and V are **not independent**.

In phase space (θ,π): $T = \frac{1}{2}g^{ij}\pi_i\pi_j$ depends on π, $V = n\mathcal{L}(\theta)$ depends on θ — **independent by construction**.

The Hamiltonian formulation is the natural one. The Lagrangian is derived from it.

Core idea

Inverting back via $\pi_i = g_{ij}\dot{\theta}^j\mathcal{L}_{\mathrm{Lag}}(\theta,\dot{\theta}) = \frac{1}{2}g_{ij}(\theta)\,\dot{\theta}^i\dot{\theta}^j - n\,\mathcal{L}(\theta)\,,\quad T - V$ **derived**, not postulated

Analogy 3 — Step 5: The Action and the Evolving Metric

From the Lagrangian (Step 4), the **action** of the training trajectory is:

$$S[\theta] = \int_a^b \mathcal{L}_{\mathrm{Lag}}(\theta, \dot{\theta})\, dt = \int_a^b \left[\frac{1}{2} g_{ij}(\theta)\, \dot{\theta}^i \dot{\theta}^j - n\, \mathcal{L}(\theta) \right] dt$$

$\delta S = 0$ selects the **classical trajectory** — the path of training given the current metric $g_{ij}(\theta)$.

Critical note

But $g_{ij}(\theta(t))$ **evolves during training** — the metric deforms as θ moves on $\mathcal{M}$.
The action S describes the trajectory **given** g_{ij}. It does not say how g_{ij} itself evolves.

Two coupled objects: the path and the geometry.

Analogy 3 Conjecture — reread from the action

There exists a reparametrisation $\tau(\theta(t))$ of training time such that the Fisher metric evolves as a Ricci flow **along the training trajectory**:

$$\frac{\partial g_{ij}(\theta)}{\partial \tau} = -2\, R_{ij}(\theta)$$

The Ricci flow is the **law of evolution of the metric** — the equation that closes the coupled system ($\delta S = 0,\ \partial_\tau g_{ij} = -2R_{ij}$).

Bonus — When $\mathcal{M}$ is Not Riemannian

The singularity problem

Neural networks have **permutation symmetries**:
swapping two neurons in the same layer gives the same model but different θ.

$\Rightarrow$ Fisher matrix is **singular**
$\Rightarrow$ $\mathcal{M}$ is not a smooth Riemannian manifold
$\Rightarrow$ it is a **stratified space** with singular strata

Consequences:

- Natural gradient is ill-defined at singularities
- Ricci flow may develop **finite-time singularities** (as in Perelman's work — handled by surgery)
- Standard Riemannian geometry breaks down

Core idea

Amari's resolution:
Use **singular learning theory** (Watanabe, 2009). Replace Fisher with a **generalised metric** on the stratified space.

The Ricci flow conjecture would need to be reformulated on this richer structure.